

2.1/2.2 Adding and Subtracting Rational Expressions - Worksheet

MCR3U

SOLUTIONS

1) Simplify and state any restrictions

a) $\frac{x+1}{18} + \frac{x-1}{45}$

$$= \frac{5(x+1)}{90} + \frac{2(x-1)}{90}$$

$$= \frac{5x+5+2x-2}{90}$$

$$= \frac{7x+3}{90}$$

b) $\frac{2}{3x} - \frac{1}{4x}$

$$= \frac{8}{12x} - \frac{3}{12x}$$

$$= \frac{5}{12x}, x \neq 0$$

c) $\frac{3}{ab} + \frac{5}{4b}$

$$= \frac{12}{4ab} + \frac{5a}{4ab}$$

$$= \frac{12+5a}{4ab}, a \neq 0, b \neq 0$$

d) $\frac{2+a}{a^2b} + \frac{4-a}{3ab^2}$

$$= \frac{3b(2+a)}{3a^2b^2} + \frac{a(4-a)}{3a^2b^2}$$

$$= \frac{6b+3ab+4a-a^2}{3a^2b^2}, a \neq 0, b \neq 0$$

2) Simplify and state the restrictions.

a) $\frac{1}{x-6} - \frac{1}{x+6}$

$$= \frac{1(x+6)}{(x-6)(x+6)} - \frac{(x-6)}{(x-6)(x+6)}$$

$$= \frac{x+6-x+6}{(x-6)(x+6)}$$

$$= \frac{12}{(x-6)(x+6)}, x \neq -6, 6$$

b) $\frac{12}{x+8} + \frac{3}{x-9}$

$$= \frac{12(x-9)}{(x+8)(x-9)} + \frac{3(x+8)}{(x+8)(x-9)}$$

$$= \frac{12x-108+3x+24}{(x+8)(x-9)}$$

$$= \frac{15x-84}{(x+8)(x-9)}, x \neq -8, 9$$

$$c) \frac{x+10}{x-6} - \frac{x-3}{x+4}$$

$$= \frac{(x+4)(x+10)}{(x+4)(x-6)} - \frac{(x-6)(x-3)}{(x+4)(x-6)}$$

$$= \frac{x^2 + 14x + 40 - (x^2 - 9x + 18)}{(x+4)(x-6)}$$

$$= \frac{23x + 22}{(x+4)(x-6)} \quad , x \neq -4, 6$$

3) Simplify and state the restrictions.

$$a) \frac{x}{x^2-9x+8} + \frac{2}{x-8}$$

$$= \frac{x}{(x-8)(x-1)} + \frac{2}{x-8}$$

$$= \frac{x}{(x-8)(x-1)} + \frac{2(x-1)}{(x-8)(x-1)}$$

$$= \frac{x + 2x - 2}{(x-8)(x-1)}$$

$$= \frac{3x-2}{(x-8)(x-1)} \quad , x \neq 8, 1$$

$$c) \frac{x}{x^2+3x+2} - \frac{3x-2}{x^2+8x+7}$$

$$= \frac{x}{(x+2)(x+1)} - \frac{3x-2}{(x+7)(x+1)}$$

$$= \frac{x(x+7)}{(x+2)(x+1)(x+7)} - \frac{(3x-2)(x+2)}{(x+2)(x+1)(x+7)}$$

$$= \frac{x^2 + 7x - (3x^2 + 4x - 4)}{(x+2)(x+1)(x+7)}$$

$$= \frac{-2x^2 + 3x + 4}{(x+2)(x+1)(x+7)} \quad , x \neq -2, -1, -7$$

$$d) \frac{x+5}{x+1} + \frac{x+2}{x-2}$$

$$\frac{(x+5)(x-2)}{(x+1)(x-2)} + \frac{(x+1)(x+2)}{(x+1)(x-2)}$$

$$= \frac{x^2 + 3x - 10}{(x+1)(x-2)} + \frac{x^2 + 3x + 2}{(x+1)(x-2)}$$

$$= \frac{2x^2 + 6x - 8}{(x+1)(x-2)}$$

$$\rightarrow \frac{2(x^2 + 3x - 4)}{(x+1)(x-2)}$$

$$= \frac{2(x+4)(x-1)}{(x+1)(x-2)} \quad , x \neq -1, 2$$

$$b) \frac{x+3}{x+5} + \frac{x+2}{x^2+3x-10}$$

$$= \frac{x+3}{x+5} + \frac{x+2}{(x+5)(x-2)}$$

$$= \frac{(x+3)(x-2)}{(x+5)(x-2)} + \frac{x+2}{(x+5)(x-2)}$$

$$= \frac{x^2 + x - 6 + x + 2}{(x+5)(x-2)}$$

$$= \frac{x^2 + 2x - 4}{(x+5)(x-2)} \quad , x \neq -5, 2$$

$$d) \frac{x+4}{x^2-121} - \frac{2x-1}{x^2+8x-33}$$

$$= \frac{x+4}{(x-11)(x+11)} - \frac{2x-1}{(x+11)(x-3)}$$

$$= \frac{(x+4)(x-3)}{(x-11)(x+11)(x-3)} - \frac{(2x-1)(x-11)}{(x-11)(x+11)(x-3)}$$

$$= \frac{x^2 + x - 12 - (2x^2 - 23x + 11)}{(x-11)(x+11)(x-3)}$$

$$= \frac{-1x^2 + 24x - 23}{(x-11)(x+11)(x-3)} \quad \rightarrow \text{see bottom of next page...}$$

4) Binomial expressions can differ by a factor of -1. Factor -1 from one of the denominators to identify the common denominator. Then, simplify each expression and state the restrictions.

$$\begin{aligned} & \frac{1}{x-2} - \frac{1}{2-x} \\ &= \frac{-1(1)}{-1(x-2)} - \frac{1}{2-x} \\ &= \frac{-1}{-x+2} - \frac{1}{2-x} \\ &= \frac{-1}{2-x} - \frac{1}{2-x} \\ &= \frac{-2}{2-x} \quad , x \neq 2 \end{aligned}$$

$$\begin{aligned} \text{b) } & \frac{2b+3}{4b-1} + \frac{b+6}{1-4b} \\ &= \frac{2b+3}{4b-1} + \frac{-1(b+6)}{-1(1-4b)} \\ &= \frac{2b+3}{4b-1} + \frac{-b-6}{-1+4b} \\ &= \frac{2b+3}{4b-1} + \frac{-b-6}{4b-1} \\ &= \frac{2b+3-b-6}{4b-1} \\ &= \frac{b-3}{4b-1} \quad , b \neq \frac{1}{4} \end{aligned}$$

$$\begin{aligned} 3 \text{ d) } &= \frac{-1(x^2-24x+23)}{(x-11)(x+1)(x-3)} \\ &= \frac{-1(x-23)(x-1)}{(x-11)(x+1)(x-3)} \quad , x \neq -1, 3, 11 \end{aligned}$$

Answers

1) a) $\frac{7x+3}{90}$, no restrictions b) $\frac{5}{12x}$, $x \neq 0$ c) $\frac{12+5a}{4ab}$, $a \neq 0, b \neq 0$ d) $\frac{6b+3ab+4a-a^2}{3a^2b^2}$, $a \neq 0, b \neq 0$

2) a) $\frac{12}{(x-6)(x+6)}$, $x \neq -6, 6$ b) $\frac{15x-84}{(x+8)(x-9)}$, $x \neq -8, 9$ c) $\frac{23x+22}{(x-6)(x+4)}$, $x \neq -4, x \neq 6$ d) $\frac{2(x+4)(x-1)}{(x+1)(x-2)}$, $x \neq -1, 2$

3) a) $\frac{3x-2}{(x-1)(x-8)}$, $x \neq 1, 8$ b) $\frac{x^2+2x-4}{(x+5)(x-2)}$, $x \neq -5, 2$ c) $\frac{-2x^2+3x+4}{(x+1)(x+2)(x+7)}$, $x \neq -7, -2, -1$ d) $\frac{-(x-23)(x-1)}{(x+11)(x-11)(x-3)}$, $x \neq -11, 3, 11$

4) a) $\frac{2}{x-2}$, $x \neq 2$ b) $\frac{b-3}{4b-1}$, $b \neq \frac{1}{4}$